

Exploring the Once in its Lifetime Gamma Ray Flare of Blazar B2 1811+31

Senior Thesis Presentation

May 2026

Cecilia Martínez Rodríguez

Physics and Astronomy Department, Barnard College



Content Outline

01

Gamma rays

Introduction to gamma-ray astronomy.

02

Cherenkov

Cherenkov radiation: atmospheric phenomena

03

VERITAS

Analysis of gamma rays via Cherenkov radiation.

04

Blazars

What are they? Why are they interesting?

05

2020 Flare

Analysis of B2 1811 + 31 observational data.

06

Takeaways

Next steps and overarching thoughts.

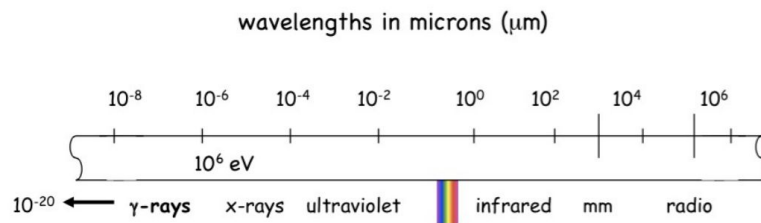
Introduction to gamma-ray astronomy

Why study gamma rays?

- Most energetic portion of the electromagnetic spectrum, spans over 8 decades of photon energy.
- Insight into a number of questions, like origin of Galactic and Extragalactic Cosmic Rays

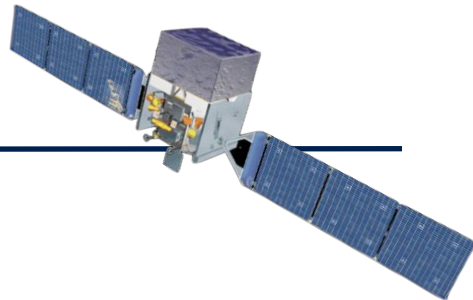
Power Sources: High Energy Cosmic Accelerators

- Electromagnetic:
 - Pulsars
- Gravitational:
 - Core collapse of massive stars
 - Accretion onto compact objects



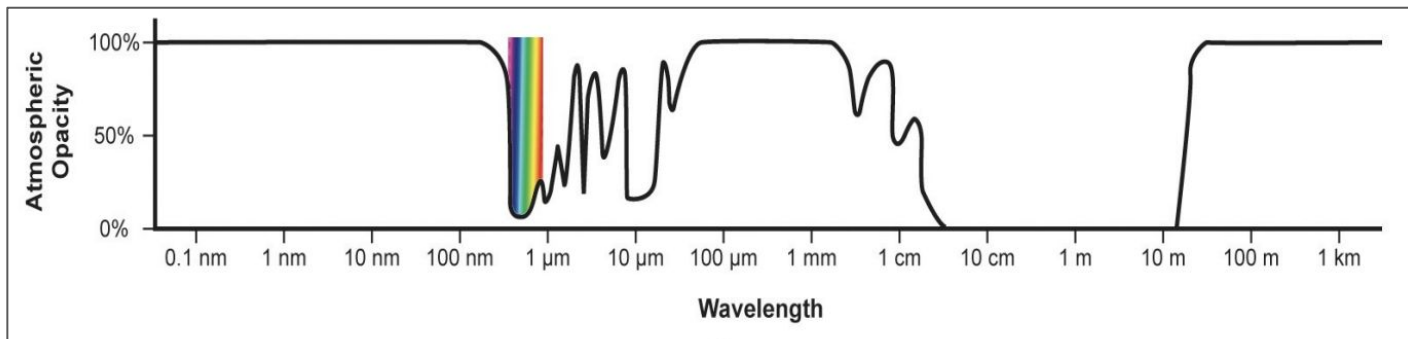
Frontiers in Astrophysics U2900, Prof. Mukherjee.

Observing gamma rays from Earth



The Challenge:

- Satellites and balloons involve sizing constraints.
 - Ex: **Fermi** surveys the gamma-ray sky in 8 keV to over 300 GeV range.
- Gamma rays, X-rays, and ultraviolet light are absorbed by the upper atmosphere.



Atmospheric opacity across the electromagnetic spectrum. Photo Credit: NASA.

The solution:

- Imaging Atmospheric Cherenkov Telescopes (IACTs)

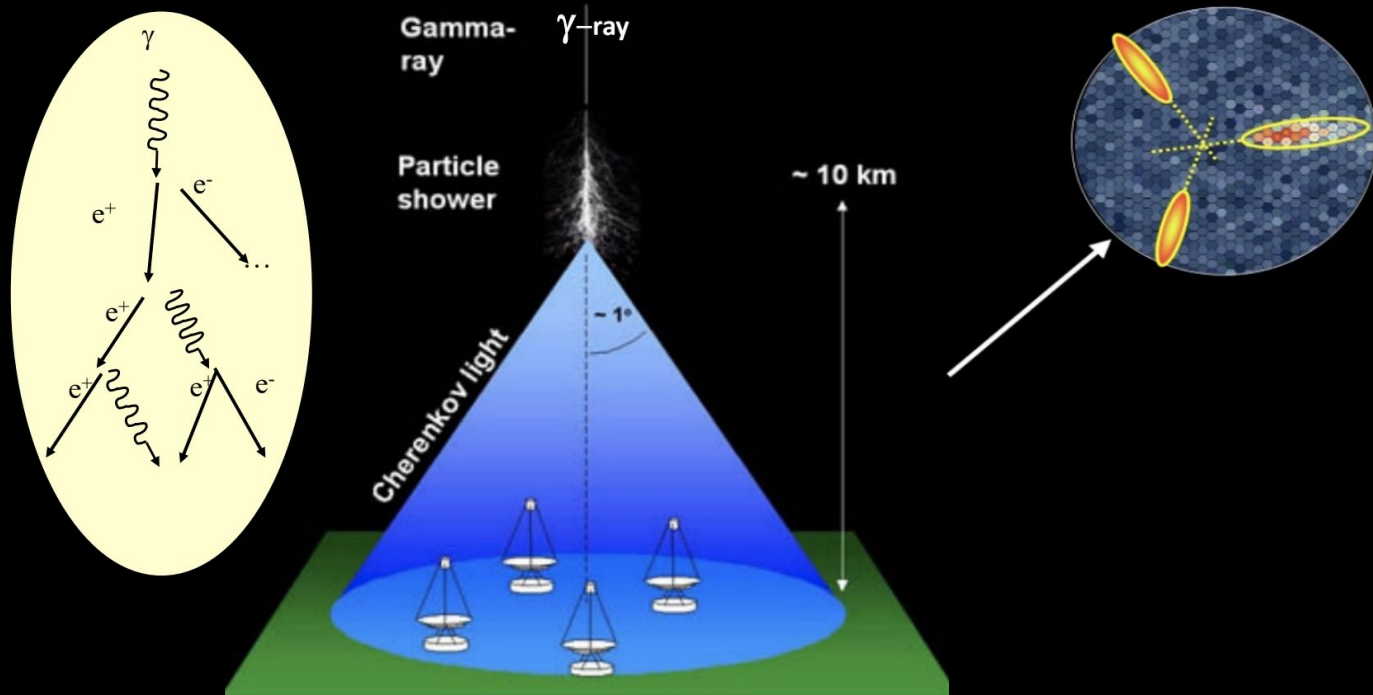


- Very Energetic Radiation Imaging Telescope Array System (VERITAS)
- Ground-based array of 4 gamma ray telescopes, each 12m wide.
- Fred Lawrence Whipple Observatory, southern Arizona
- Very High Energy (VHE) ~ 85 GeV - >30 TeV
- Part of multiwavelength & multimessenger collaborations



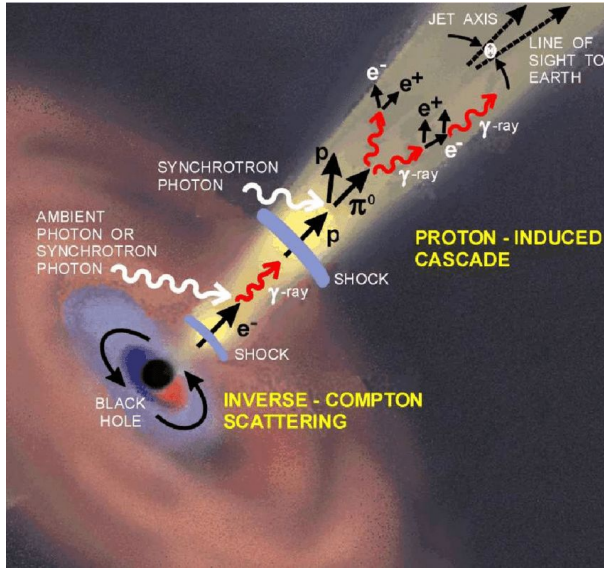
The VERITAS Array (Credit: CfA | Harvard & Smithsonian)

The Atmospheric Cherenkov Technique



Modified from Frontiers in Astrophysics U2900, Prof. Mukherjee.

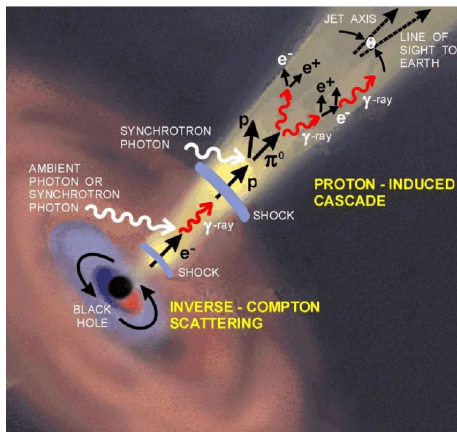
Introduction to Blazars



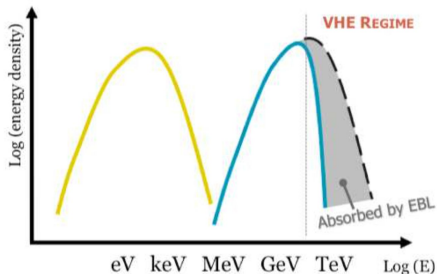
Representation of gamma ray production in relativistic jets.
(Buckley J 1998.)

- **Active Galactic Nuclei** → bright central regions of supermassive black holes (SMBHs) that are actively accreting mass
- **Blazars** → AGN with jets of highly energetic plasma pointed ~ directly at Earth
- **Jets** → the strong, twisting magnetic field around the polar regions of the SMBH **accelerates** & **collimates** charged particles.
 - Different acceleration mechanisms can be observed at different wavelengths.

Spectral Energy Distributions



Representation of gamma ray production in relativistic jets.
(Buckley J 1998.)



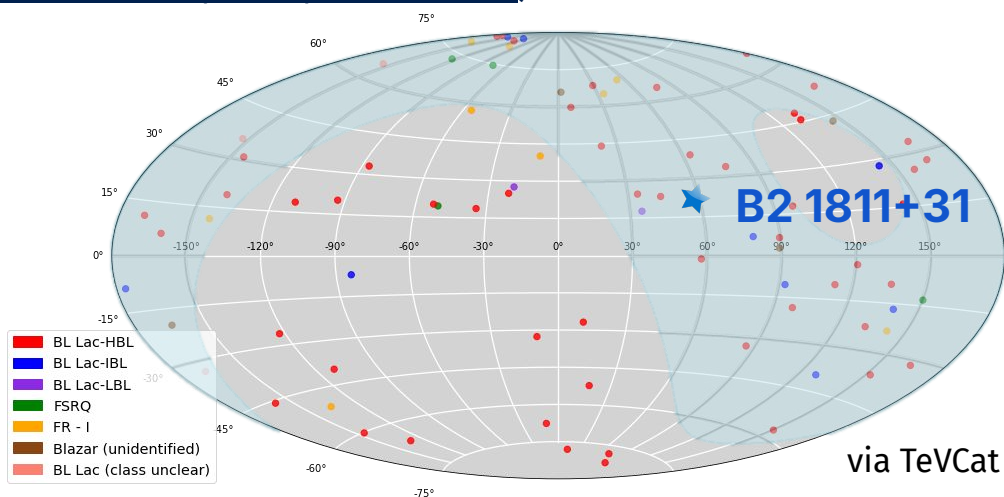
General Blazar SED (Prandani 2010)

Spectral energy distributions (SEDs) characterize the configuration and processes driving emission. Jet acceleration mechanisms are divided into two main components:

- 1) The Lower Energies (Radio → X-ray)
 - **Synchrotron Emission:** charged particles (electrons) accelerated in a curved path by the magnetic field
- 2) The Higher Energies (X-ray - gamma-ray)
 - **Leptonic Models: Inverse Compton Scattering**
 - Particles "bump" into lower-energy photons, boosting them into the gamma ray regime.
 - **Hadronic Models:**
 - Ex: proton synchrotron

B2 1811+31

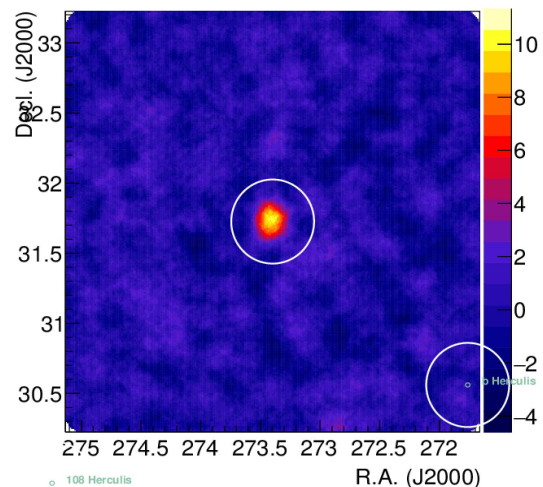
- B2 1811+31 (R.A.=18 13 35.2027 Dec.=+31 44 17.620, J2000.0) is a blazar located at redshift of $z=0.117$ (Giommi et al. 1991).
 - Like most blazars, first detected in a lower energy range, as a radio emitter.
 - Classified as an Intermediate Synchrotron Peaked BL Lacertae (IBL) (Laurent-Muehleisen, S. A., et al. 1999).



First significant detection by VERITAS

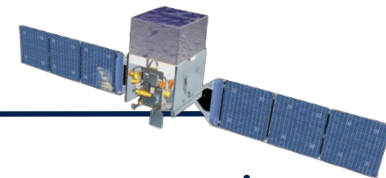
- Regularly not detected by VERITAS
 - 0.18 σ in 3.38 hrs
- In October of 2020, Fermi and MAGIC issued an alert reporting to possible elevated state via The Astronomer's Telegram (ATel).
- Observed by VERITAS in **VHE** for the first time:
 - **October 15-19, 2020**
 - **4.33** hours of data
 - **Significance:** 9.18 σ
- Data processed using VEGAS (VERITAS Gamma-ray Analysis Suite)

Significance Map (smoothed)

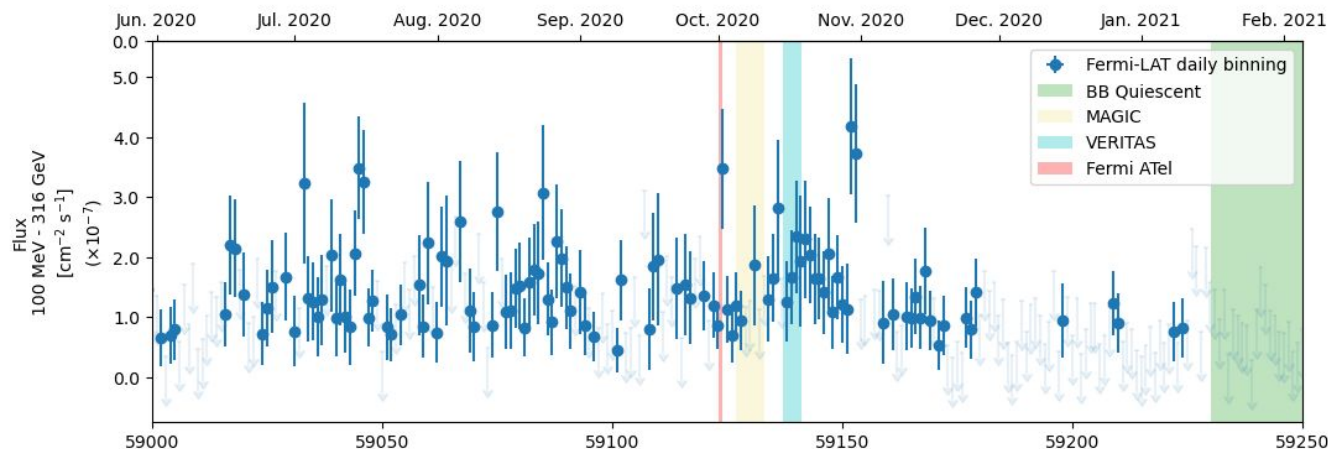


VERITAS Sky Map: Flare Detection

Characterizing the Flare Period

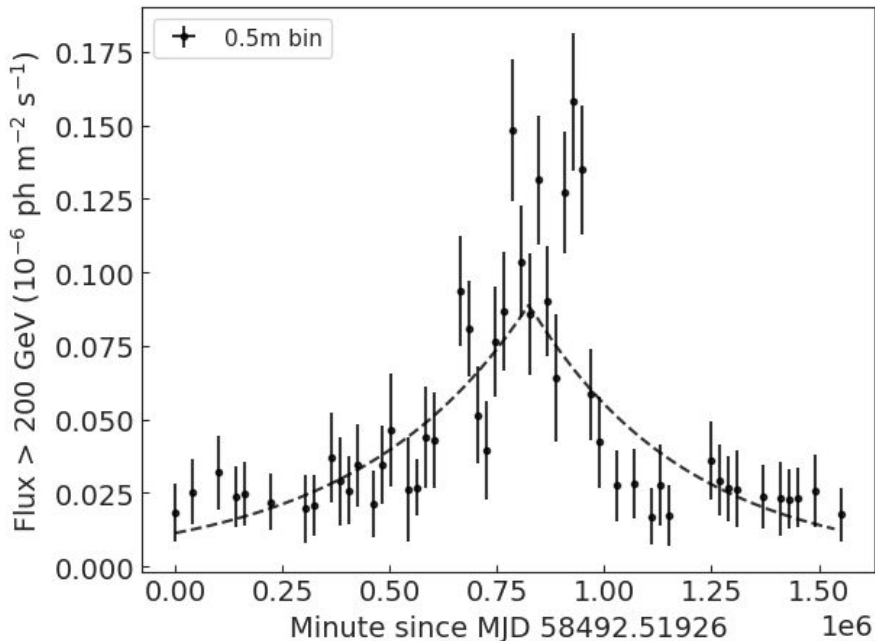
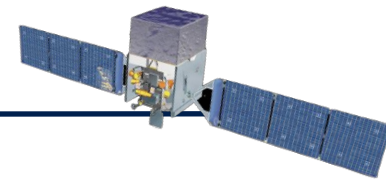


- When VERITAS followed up on the ATel alert (Oct 15-19, 2020), we saw a rise in energy flux across the first day, then a decay.



- When did the elevated flux period begin? When did it end?

Need for Time-Scale Analysis

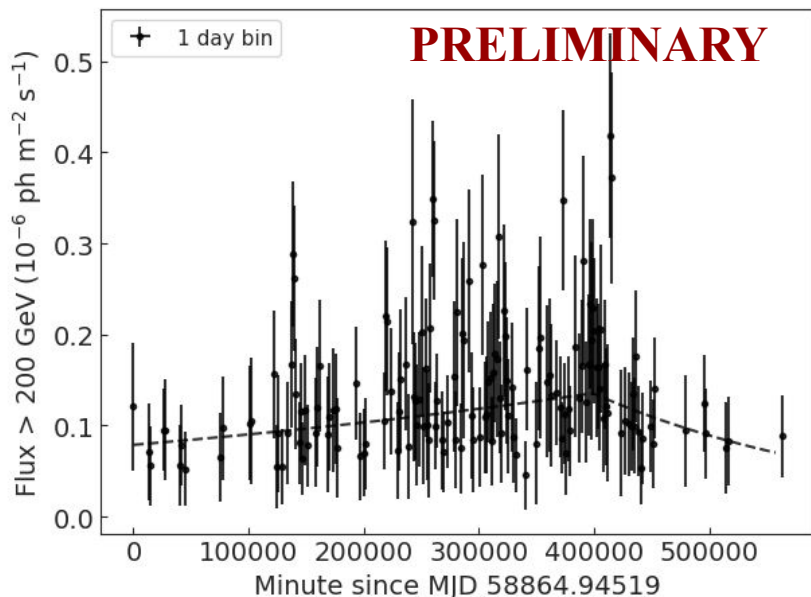


Rise and Decay Analysis of Fermi Lightcurve

- Rise and Decay analysis at half-month bin scale for a period of two years surrounding the flare event.
- **Goal:** Understanding the alert timing in October. Was it one or multiple peaks?

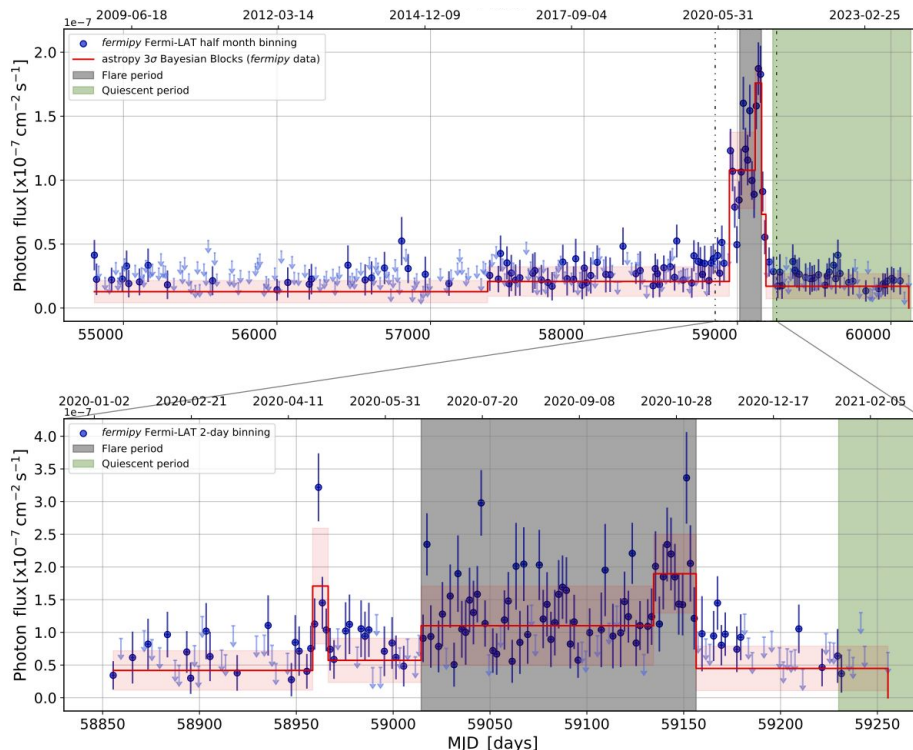
Need for Time-Scale Analysis

Fermi-LAT 1 day bins, Jan 2020 (58864) - Feb 2021 (59255)



- Possibility of multiple mini-flares more noticeable at 1-day binned scale for a year surrounding the flare alert.
- Does the flux point distribution reveal anything of rise/decay timescales?

Lightcurve Analysis of Flare Period

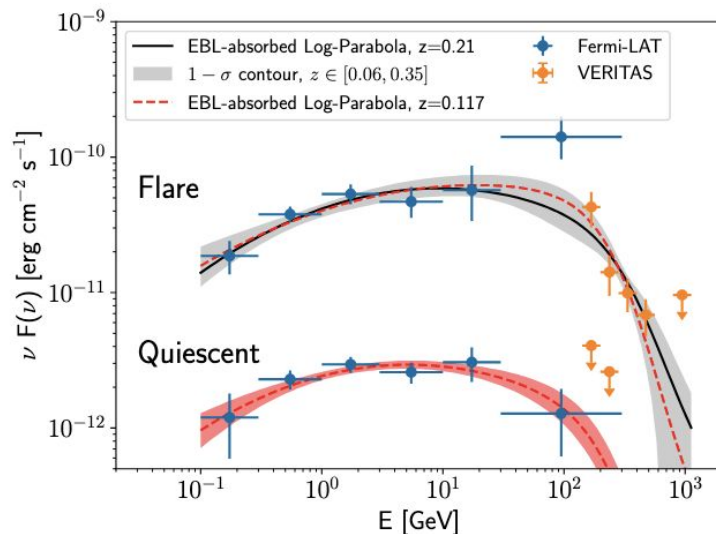


Analysis of Fermi lightcurve (P.Drake).

- Relatively stable baseline sustained for a decade (2009-2019).
- During 2020, elevated emission for a period of ~6 months.
- Return to baseline after 6 months.



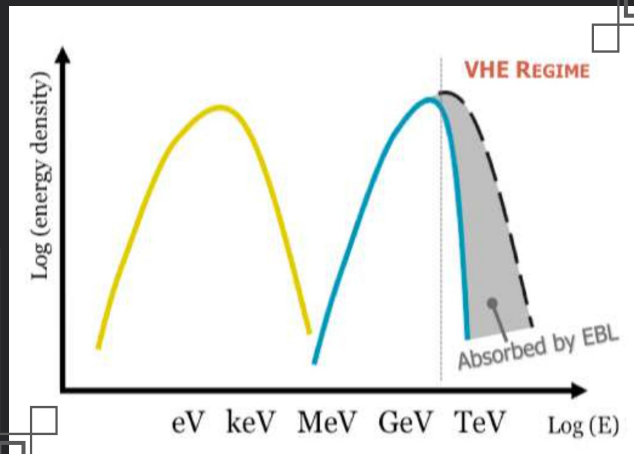
Gamma-Ray Spectral Energy Distribution



B2 1811+31 Gamma-ray SED during both quiescent and flaring period (paper in progress). EBL model from Saldana-Lopez et al. (2021).

- **Extragalactic Background Light (EBL)** scatters and absorbs light, dimming the observed flux. We account for it with an absorption model.
- Using multiwavelength data from Fermi and VERITAS, we plotted the SED of both the quiescent and flaring periods.
- Both curves, despite the difference in flux, fit best to the same model.
 - **Log-Parabola** → implies non-thermal, stochastically accelerated particles

Spectral Models



(Prandani 2010)

Extragalactic Background Light (EBL) scatters and absorbs light, **dimming the observed flux**, so we account for absorption in all fits. To compare the fitting among models with different parameters, we use Bayesian Information Criterion (BIC). The closer the BIC are to 0, the better the fit.

- ★ **Power law (PWL)**: slow decline in flux.
- ★ **Exponential cutoff power law (EXPWL)**: exponential decay at higher energy cutoff.
- ★ **Log parabola (LP)**: curve and soft drop.

Currently, the **flare** is best fit to an **LP** and the **quiescent** to an **EXPWL** with a 3σ threshold.

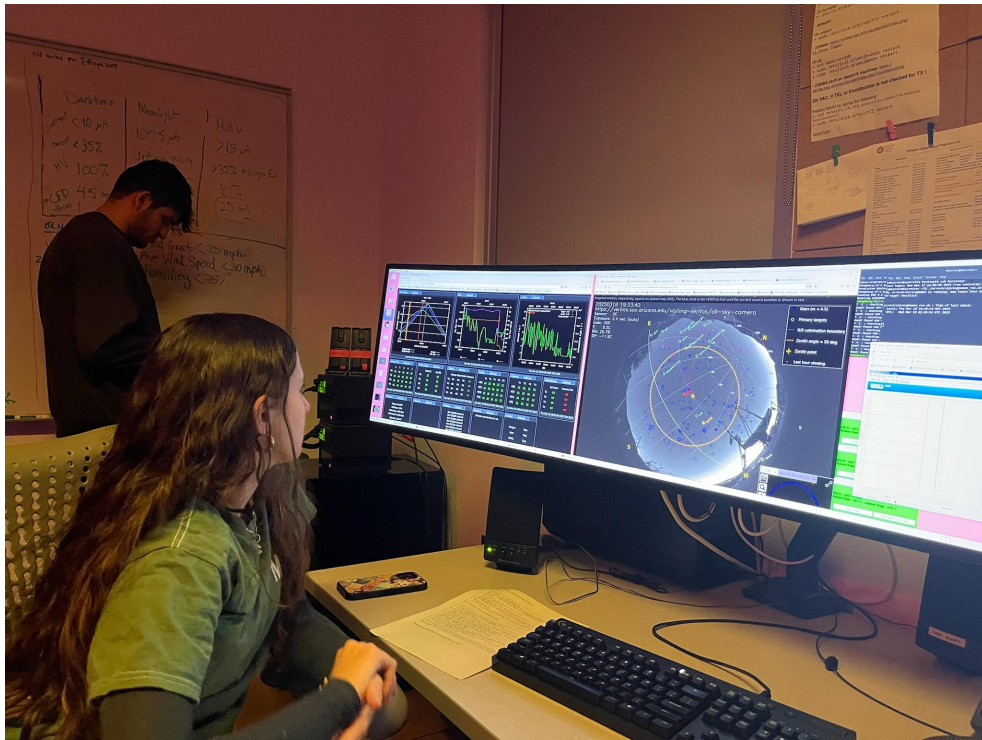
EBL-Absorbed Log-Parabola Model & Parameters

$$\frac{dN}{dE} = N_0 \left(\frac{E}{E_0} \right)^{-\Gamma - \beta \log_{10}(\frac{E}{E_0})} e^{-\tau(E,z)}$$

Table 1. Parameters of the spectral models fitted to the *Fermi*-LAT and VERITAS combined spectra.

Parameters	Quiescent Period		Flaring Period	
	$z = 0.117$	$z = 0.117$	$z = 0.21$	
N_0 [erg cm ⁻² s ⁻¹]	$(2.94 \pm 0.21) \cdot 10^{-12}$	$(5.74 \pm 0.96) \cdot 10^{-11}$	$(5.88 \pm 0.94) \cdot 10^{-11}$	
Γ	1.987 ± 0.045	1.932 ± 0.034	1.82 ± 0.034	
β	0.076 ± 0.022	0.083 ± 0.018	0.070 ± 0.017	
E_0 [GeV]	4.1	9.1	9.1	

VERITAS @ work!



Takeaways & Next Steps

- To try to understand this blazar's variability, we aim to produce and fit a broadband SED of both the flaring and quiescent states
 - Multi-component emission, i.e., spine-sheath or similar models.
- Continued monitoring of B2 1811+31
 - An additional flare would allow for clarity, allowing us to better understand whether this was a unique event, or a potentially cyclic component.
- Constraining multiwavelength time scales
 - We have currently constrained the observed Fermi-LAT time scale, but if different parts of the spectrum are powered by different mechanisms in the jet, we might expect a different time-scale in the X-ray or radio.

Thank you!

Thanks to Prof. Mukherjee for the invaluable guidance throughout my four years here at Barnard. Thanks to the amazing group of people involved in the VERITAS collaboration, Matthew Lundy in preparation for today, and thanks to the Astronomy and Physics Department for fostering such a positive learning environment.

VERITAS research at Barnard is supported by grants from the NSF.



References

Drake, P., & Adams, C. (2023). Veritas follow-up observation of the BL Lac Blazar B2 1811+31 2020 flare. Proceedings of 38th International Cosmic Ray Conference — PoS(ICRC2023). <https://doi.org/10.22323/1.444.0701>

Feng, Qi. “Exploring the High-Energy Gamma-Ray Spectra of TEV Blazars.” Proceedings of 37th International Cosmic Ray Conference — PoS(ICRC2021), 2021, doi:10.22323/1.395.0802.

Giommi, P., Tagliaferri, G., Beuermann, K., et al. 1991, The Astrophysical Journal, 378, 77, doi: 10.1086/170408

Hanxiao Xia et al 2026 ApJ 1001 32, doi: 10.3847/1538-4357/ae4dde

Marek Sikora, Mieszko Rutkowski, Mitchell C. Begelman, A spine–sheath model for strong-line blazars, Monthly Notices of the Royal Astronomical Society, Volume 457, Issue 2, 01 April 2016, Pages 1352–1358, <https://doi.org/10.1093/mnras/stw107>

Meyer, E. T., Fossati, G., Georganopoulos, M., & Lister, M. L. (2011). From the blazar sequence to the blazar envelope: Revisiting the relativistic jet dichotomy in radio-loud active galactic nuclei. The Astrophysical Journal, 740(2), 98. <https://doi.org/10.1088/0004-637x/740/2/98>

Meyer, Manuel, et al. “Characterizing the Gamma-Ray Variability of the Brightest Flat Spectrum Radio Quasars Observed with the Fermi Lat.” The Astrophysical Journal, vol. 877, no. 1, 2019, p. 39, doi:10.3847/1538-4357/ab1651.

References

Park, N. 2016, in Proceedings of The 34th International Cosmic Ray Conference — PoS(ICRC2015) (The Hague, The Netherlands: Sissa Medialab), 771. <https://pos.sissa.it/236/771>

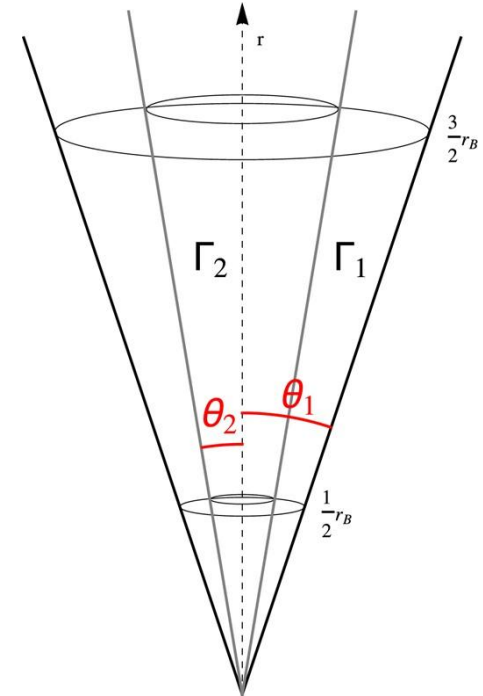
Prandini, E., Bonnoli, G., Maraschi, L., Mariotti, M., & Tavecchio, F. (2010). Constraining blazar distances with combined Fermi and TEV Data: An empirical approach. *Monthly Notices of the Royal Astronomical Society: Letters*, 405(1). <https://doi.org/10.1111/j.1745-3933.2010.00862.x>

Scargle, J. D. (1998). Studies in astronomical time series analysis. V. Bayesian blocks, a new method to analyze structure in photon counting data. *The Astrophysical Journal*, 504(1), 405–418. <https://doi.org/10.1086/306064>

Additional Materials

Spine-Sheath Model

- Potential to understand flare duration in spine (high Lorentz factor, strong collimation) and sheath (lower Lorentz factor, exterior) interactions.
- In addition to expected radiation sources, external-synchrotron Compton" (ESC) emission might occur.
 - Internal interactions between spine and sheath may lead to shocks or variabilities that accelerate particles.



M. Sikora, et al.